Empirical Investigation of the Taylor Rule and Interest Rate Dynamics

Macroeconomic Model Selection and Diagnostic Testing using FRED Dataset

Section (a): General-to-Specific Model Selection Approach

To construct a parsimonious, statistically optimal model for the Federal Funds Rate (**INTRATE**), we begin with an Unrestricted Linear Model containing all 7 macroeconomic indicators available in the FRED dataset (1960–2014):

$$INTRATE_t = \beta_0 + \beta_1 INFL_t + \beta_2 PROD_t + \beta_3 UNEMPL_t + \beta_4 COMMPRI_t + \beta_5 PCE_t + \beta_6 PERSINC_t + \beta_7 HOUST_t + \varepsilon_t$$

The OLS multi-variable base estimation gives an initial R^2 of 0.6385, an AIC of 2916.30, and a BIC of 2956.73. We apply a backward elimination process at a 5% significance rule:

Step 1: Identify **UNEMPL** as the least significant variable with a p-value of 0.2899 (t-stat = 1.059). Removing it yields an intermediate model with R^2 = 0.6379, AIC = 2915.44, and BIC = 2951.37.

Step 2: In the reduced configuration, **PROD** becomes the least significant parameter with a high p-value of 0.3232 (t-stat = -0.989). Dropping PROD yields the final parsimonious model.

Step 3: All remaining predictors exhibit t-statistics well above the absolute value of 2.0 and p-values strictly below 0.05. The elimination process terminates successfully.

Final Parsimonious Equation:

$$INTRATE_t_hat = -0.2401 + 0.7175\ INFL_t - 0.0075\ COMMPRI_t + 0.3405\ PCE_t + 0.2402\ PERSINC_t - 0.0205\ HOUST_t$$

Economic Grounding: The strong positive parameters on Inflation (0.7175) and Consumption Expenditure (0.3405) conform strictly to macroeconomic theory; the Federal Reserve aggressively raises interest rates when inflation or demand accelerates to stabilize market conditions.

Section (b): Specific-to-General Model Selection Approach

We reverse the framework using a Forward Selection routine starting with a constant baseline configuration. Variables are introduced sequentially based on maximum AIC minimization criteria. **INFL** is introduced first, followed by **PERSINC**, **PCE**, **HOUST**, and **COMMPRI**. The inclusion of either UNEMPL or PROD increases the absolute information criteria.

Conclusion: Both the forward and backward routines converge on the exact same 5 variables with matching magnitudes. This path-independence guarantees that our parsimonious specifications are structurally robust.

Section (c): Structural Model Comparison Table

We directly benchmark our multi-variable model selected in parts (a/b) against the classical simplified two-variable Taylor Rule:

Evaluation Metric	Selected Parsimonious Model	Classical Taylor Rule Model
Variables Contained	INFL, COMMPRI, PCE, PERSINC, HOUST	INFL, PROD
Multiple R^2	0.6374	0.5747
Adjusted R^2	0.6346	0.5734
Akaike Info Crit. (AIC)	2914.42	3013.62
Bayesian Info Crit. (BIC)	2945.87	3031.59

Section (d): Diagnostics, Breakpoint Analysis, and Forecasting

1. Ramsey RESET Test: The test yields a statistic of F = 2.5371 (p-value = 0.1117). We fail to reject the null hypothesis at the 5% level, establishing that the structural linear form is properly specified.

2. Chow Structural Breakpoint Test (January 1980): The full dataset is partitioned into sub-periods before and after the 1980 boundary. The test generates an F-statistic of 28.735 (p-value < 0.001), indicating an overwhelming rejection of parameter stability across time.

3. Jarque-Bera Residual Analysis: The test rejects error normality (χ^2 = 12.444, p-value = 0.00198), signaling volatility clustering and outliers in interest rate regimes.

4. Out-of-Sample Forecasting (2000:01 to 2004:12): Evaluation of predictive accuracy via the Chow Forecast Test (F = 5.511, p-value < 0.001) confirms structural breakdown out-of-sample. The baseline Taylor Rule displays systematic forecast errors during this period because it fails to capture the Federal Reserve's historical shift toward keeping interest rates abnormally low to offset the post-dot-com recession shocks.

Policy Insights: The January 1980 break marks the historic transition to the *'Volcker Regime'* under Fed Chairman Paul Volcker. Confronted with chronic stagflation, the Fed restructured its operational rule to prioritize monetary supply aggregates over interest rate damping. Consequently, split sub-sample estimation is required to maximize practical policy forecasting precision.